Mars Global Mosaic: Mapping Lava Flows, Fluvial Channels, and Impact Craters in Visible and Infrared

REMOTE SENSING 2026 OCN 4704 · Logan Edwards

Interim Report · status as of 27 October 2026 (results to 26 October)

1. Project Investigators

Logan Edwards, REMOTE SENSING 2026 OCN 4704.

2. Project Goals

Assemble one co-registered visible and thermal-infrared mosaic of Mars in ArcGIS Pro, over the ±60° band the THEMIS night mosaic covers (86.6% of the surface), and use it to map lava flows, fluvial channels and impact craters.

- Put Viking visible colour, THEMIS day and night infrared and HRSC/MOLA topography on one grid over ±60°.
- Map the three landform families from that stack, first in a type area (Ius Chasma), then across ±60°.
- Test whether thermal infrared separates surfaces that visible light cannot.
- Score every product against something it never saw: held-out hand labels, the IAU crater gazetteer and the USGS geologic map.

3. Relevant Spectral Bands

Band or product	Wavelength	Cell	What it contributes
Viking MDIM 2.1 R, G, B	visible, ~0.4–0.7 µm	232 m	Colour and shading: bright dust, dark basalt. High-pass filtered at ~50 km; the colour, an uncalibrated blend, still carries regional differences.
THEMIS Day IR (v12)	thermal IR, 12.57 µm	100 m	Daytime temperature. Ground that is bright in visible stays cooler by day (r −0.49 at Ius).
THEMIS Night IR (v14, ±60°)	thermal IR, 12.57 µm	100 m	Night temperature: rock holds heat. The band least like the others, $\|r\| \leq 0.12$.
HRSC/MOLA blended DEM	altimetry + stereo	200 m	Slope in degrees, local relief, hillshade, flow routing, closed-depression depth.
Diurnal contrast index	day − night, relative	100 m	Separates materials inside one area; not thermal inertia, not comparable across the planet.

Correlation between the five image bands at Ius Chasma:

r	Viking R	Viking G	Viking B	Day IR	Night IR
Viking R	1	+0.960	+0.925	−0.486	−0.121
Viking G	+0.960	1	+0.993	−0.496	−0.085
Viking B	+0.925	+0.993	1	−0.488	−0.068
Day IR	−0.486	−0.496	−0.488	1	+0.105
Night IR	−0.121	−0.085	−0.068	+0.105	1

The three Viking bands behave as one (r 0.93–0.99); day IR adds a second dimension and night IR a third. That is the measured case for fusing them.

4. Project Tasks and Percent Complete

#	Task	Where it stands	13 Sep	Now
1	Scope, targets and type area	Two type areas, Ius Chasma and Athabasca Valles; the ±60° extent	90%	100%
2	Raster acquisition	Three global mosaics plus the THEMIS night IR mosaic	100%	100%
3	Statistics, stretches, pyramids	Done for every product; pyramids on the four source globals took 26 min	70%	100%
4	Grouped mosaic viewer	Done	100%	100%
5	DEM terrain derivatives	Slope rebuilt in degrees for the type area and ±60°	80%	100%
6	Image-gradient products	Slope run on imagery, not the DEM; relabelled as gradient and not used	70%	100%
7	Jezero context	Hosted services only; the dead HiRISE link removed	20%	100%
8	Classification	±60° SVM on hand-drawn labels, 200 m: 73.8%, κ 0.58 held out; 971 seven-class labels applied, maps running	20%	80%
9	CRS harmonisation	Done: one 100 m / 200 m grid over ±60°	15%	100%
10	Visible + IR composite	Done: type area; 7-band ±60° stack	5%	100%
11	Landform digitising	Review begun at Ius: 94 channels and 26 crater rims accepted; Athabasca and lava margins to do	0%	20%
12	Crater inventory	Robbins catalogue in the project (385,049); 465,354 closed depressions ≥ 1 km at ±60°	0%	60%
13	Map layouts	17 map layouts (none in September)	0%	85%
14	Report and presentations	Prospectus and Presentation 1 delivered; interim built to the 26 Oct cut; final paper and deck drafted	25%	60%

5. Processing Completed Since the Last Report

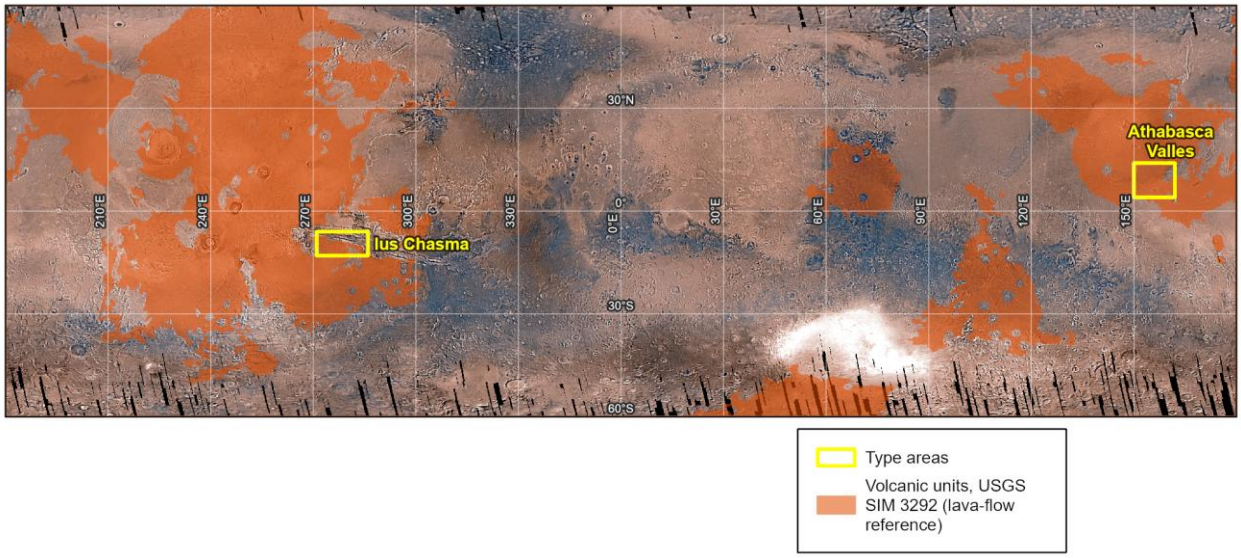
When	What	Outcome
18 Sep	Night IR acquired; type-area stack; slope on a metric grid	Four inputs on one 100 m grid, 100% co-valid; slope in degrees
19 Sep	±60° grids and diurnal contrast index	15.2 bn px in 36.3 min
19 Sep	Channel and crater candidates, Ius; basins at ±60°	2,610 channels; 1,685 closed depressions; 5,144 basins ≥ 20 km
24–30 Sep	Global composites built in ArcGIS Pro	Two runs finished: 2 h 52 m and 3 h 10 m
29 Sep	Hand-drawn training polygons; SVMs	Four classes; two ±60° maps
2–3 Oct	First SVM maps checked; corrected stack scored held out	Tile artefact, mostly elevation; then 73.5%, κ 0.58
6–7 Oct	Backup to GitHub; class codes	Public repository; pixel = class code
8 Oct	Reference data, Athabasca, tests T1–T3 and T8	Thermal adds +1.3 points; ten layouts
8–9 Oct	Digitising by review; tests T4–T7; validation sheet	No support for H1–H3 or a tributary count; 12 layouts

When	What	Outcome
13–15 Oct	Pyramids; ±60° craters and channels ≥ 1 km; 180° DEM fix	465,354 closed depressions, 969,822 channel segments
14–16 Oct	200 m classification; T1 with Pro's SVM; the published map	73.8%, κ 0.58; thermal adds +1.36 and +1.97 points
19–21 Oct	Deep learning: a box detector, then a U-Net over ±60°	U-Net 98.9%, κ 0.98 held out, as terrain provinces
22–27 Oct	The 512 labels audited; seven classes proposed and reviewed	Five Crater labels are volcanoes; 971 seven-class labels applied

6. Preliminary Results

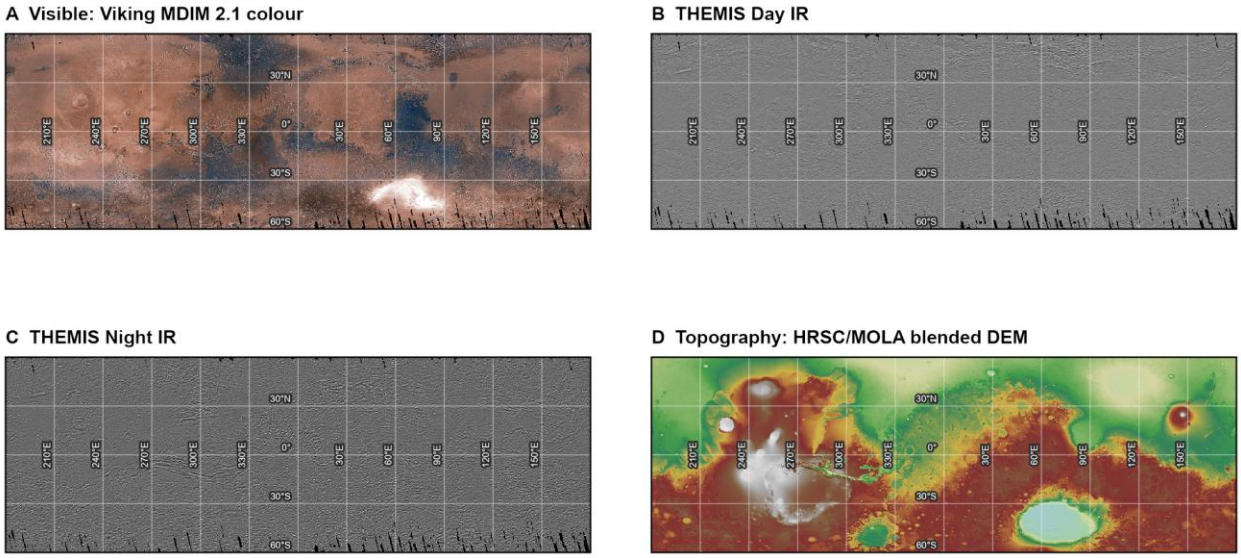
Every map is a layout in the ArcGIS project, exported for this report.

Figure 1. The analysis extent and the two type areas



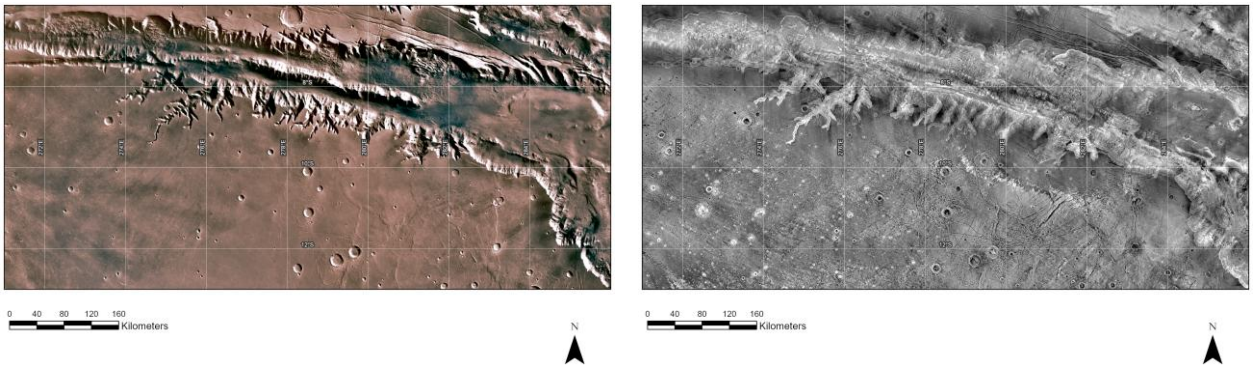
- The analysis covers ±60° latitude, 86.6% of the planet: the THEMIS night mosaic's coverage
- Two type areas: Ius Chasma (canyon and tributaries) and Athabasca Valles (flood lava)
- Orange: the USGS geologic map's volcanic units, used as the lava-flow reference

Figure 2. The mosaic at ±60°



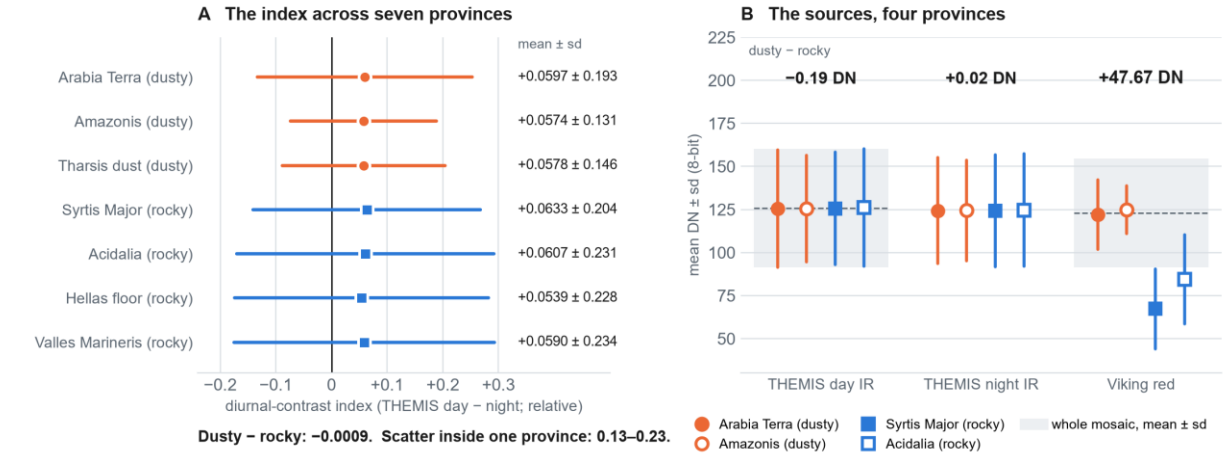
- All four datasets on one 200 m grid
- Composite Bands failed four times on the global rasters; projecting first and bounding the extent fixed it

Figure 3. Ius Chasma type area: visible and night IR



- Left, Viking visible colour; right, THEMIS night IR. 271–286°E, 6–13°S: canyon walls, plateau, Louros Valles
- Same 100 m grid, every pixel valid in all five bands
- Night IR barely correlates with any other band ($|r| \leq 0.12$), so it adds information the others lack

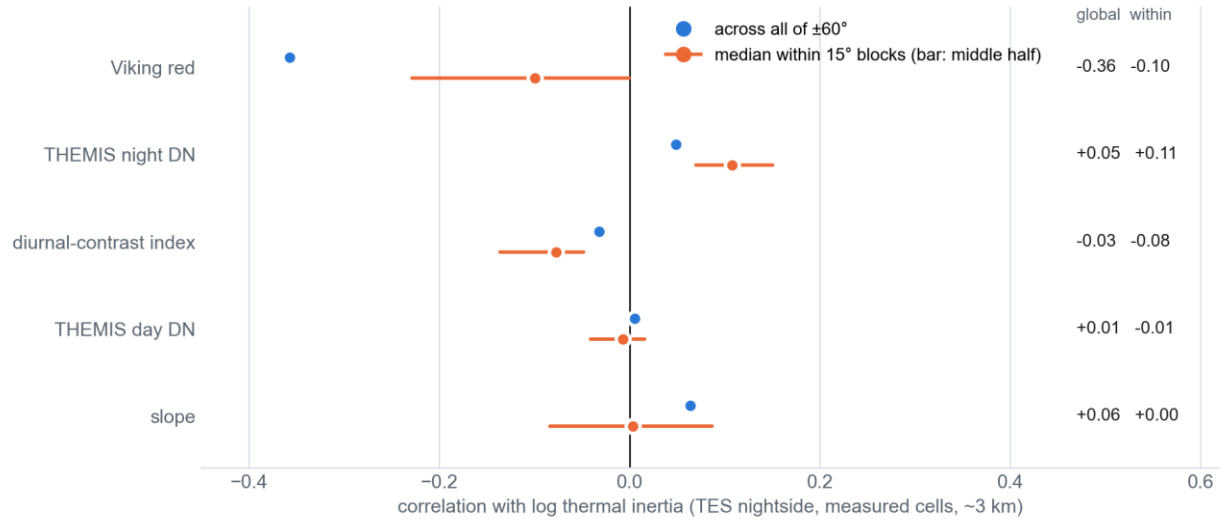
Figure 4. Can the thermal pair map material across ±60°?



Both THEMIS mosaics are 8-bit DN stretched region by region: each province repeats the mosaic-wide mean and spread, so the index is local contrast, not thermal inertia. Viking is not stretched that way and keeps the regional difference in its colour. Dusty − rocky in B: the provinces' group means. No whole-mosaic mean is recorded for night IR.

- Viking red separates the dusty provinces from the rocky ones by 47.67 DN
- THEMIS day and night IR separate them by −0.19 and +0.02 DN; the index, over seven provinces, by −0.0009
- Every province repeats the whole mosaic's mean and spread, because both THEMIS mosaics were stretched region by region
- Across the planet, the material signal is in Viking colour

Figure 5. Checked against calibrated thermal inertia (TES)



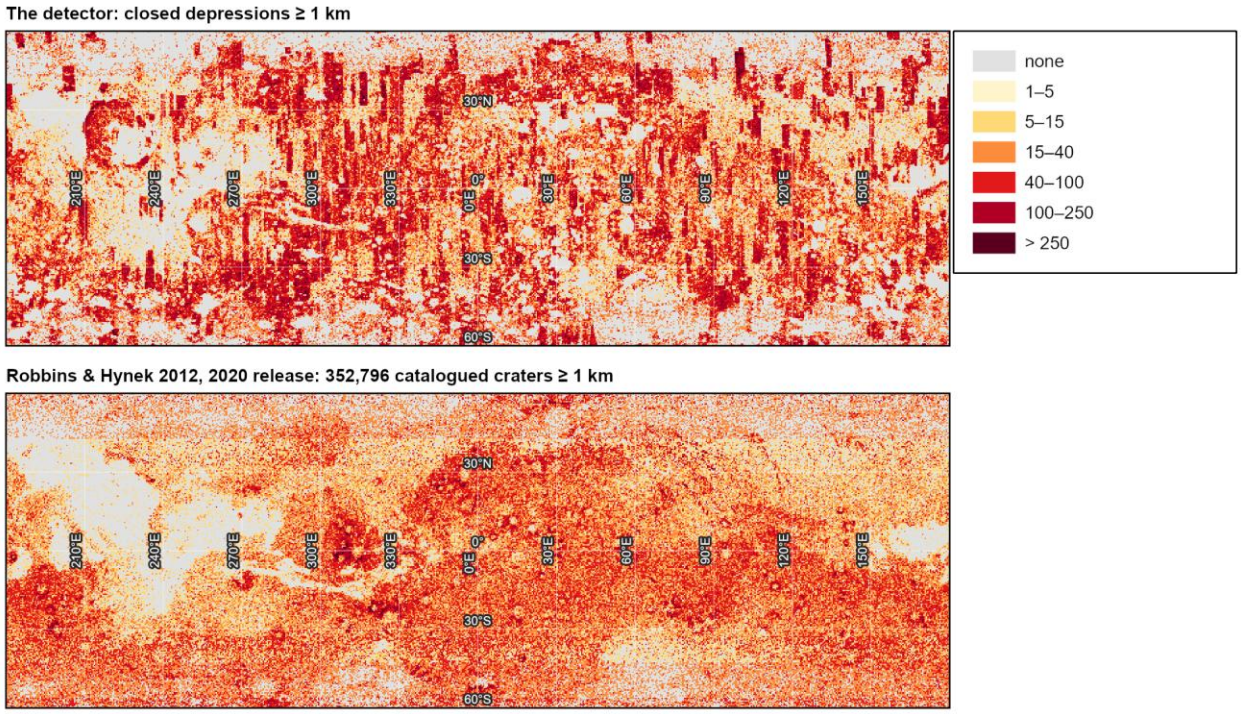
- Viking colour follows calibrated thermal inertia across the planet (r −0.36)
- Night IR and the diurnal-contrast index follow it weakly, and only inside regions (|r| ≈ 0.1)
- Day IR does not follow it, and across the planet neither THEMIS mosaic does
- At TES's ~3 km cells the material signal is in Viking colour; thermal IR adds little

Figure 6. Craters at ±60°: closed depressions ≥ 20 km



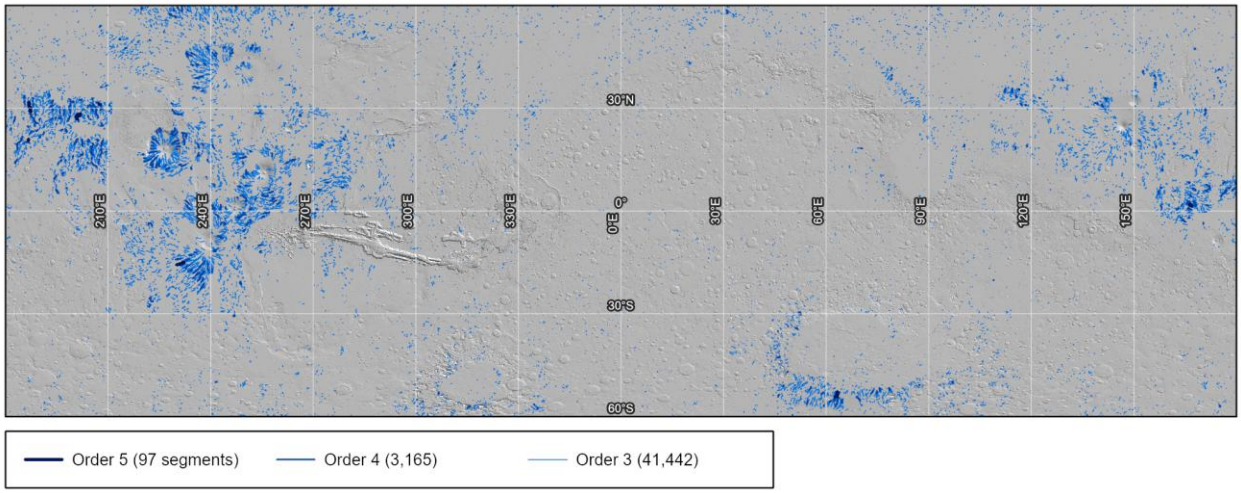
- 5,144 basins found by fill depth in 48 s
- 68% of 117 IAU craters ≥ 100 km recovered, diameter within ±50%
- Median diameter error +2.2%
- Breached craters are missed by design: they do not close

Figure 7. Closed depressions ≥ 1 km across ±60°



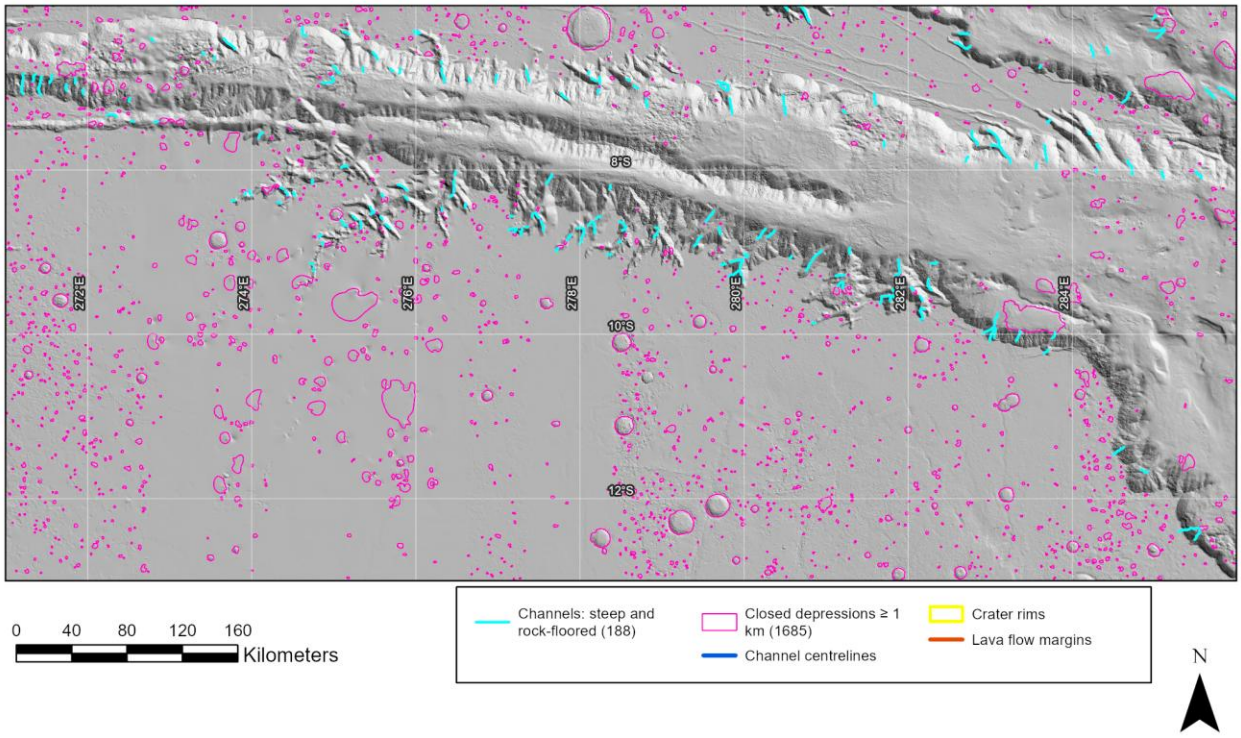
- 465,354 closed depressions ≥ 1 km, from an 11-hour run
- Against Robbins: 15.3% are catalogued craters at ≥ 1 km, 62.2% at ≥ 5 km, 83.1% at ≥ 20 km
- Their density per 10^4 km² barely follows the catalogued craters (r −0.11); below ~5 km it follows the DEM's sharpness
- Good for large craters and as prompts for review; for crater counts, Robbins is the reference

Figure 8. Channels across ±60°



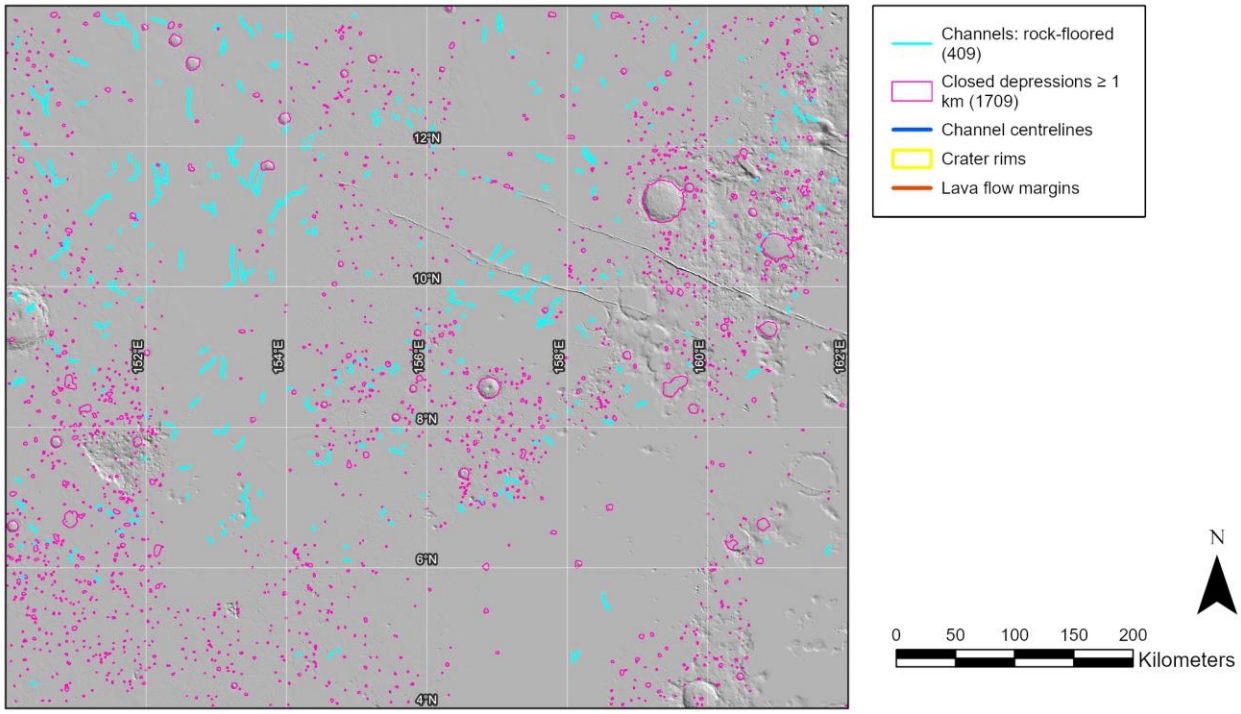
- 969,822 routed channel segments; orders 3–5 drawn
- 60.5% of stream cells discarded as fill artefact (Ius alone: 56%)
- 89.8% lie on ground under 2°, so the plateau channels are DEM noise
- No channel reference exists at ±60°, so the network is not scored

Figure 9. Ius Chasma: candidates to digitise



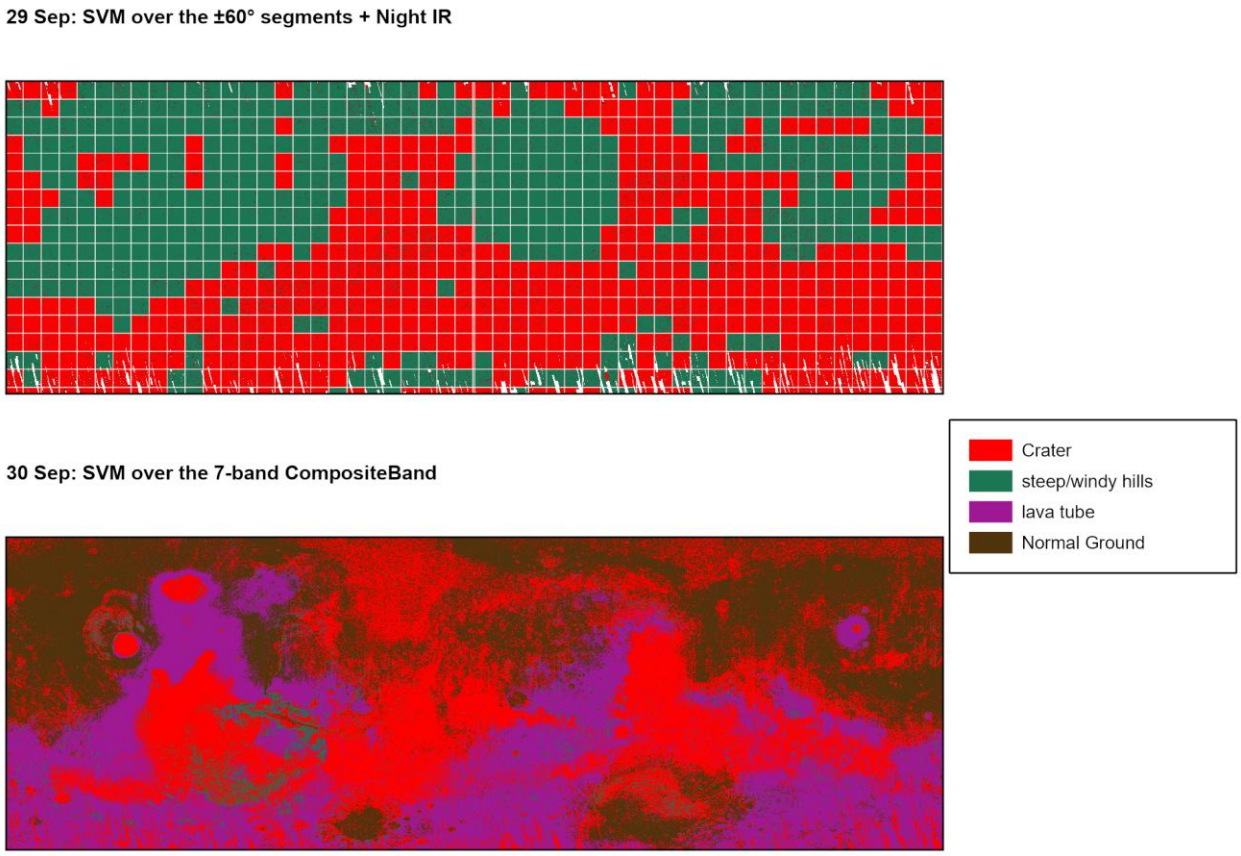
- Fill raised the canyon floor by up to 2,077 m; 56% of the first channel network was artefact
- 188 channel candidates are steep and rock-floored; reviewed first, 94 accepted
- Only 11% of the 1,685 closed depressions ≥ 1 km match a Robbins crater, so they are prompts for review

Figure 10. Athabasca Valles: the second type area



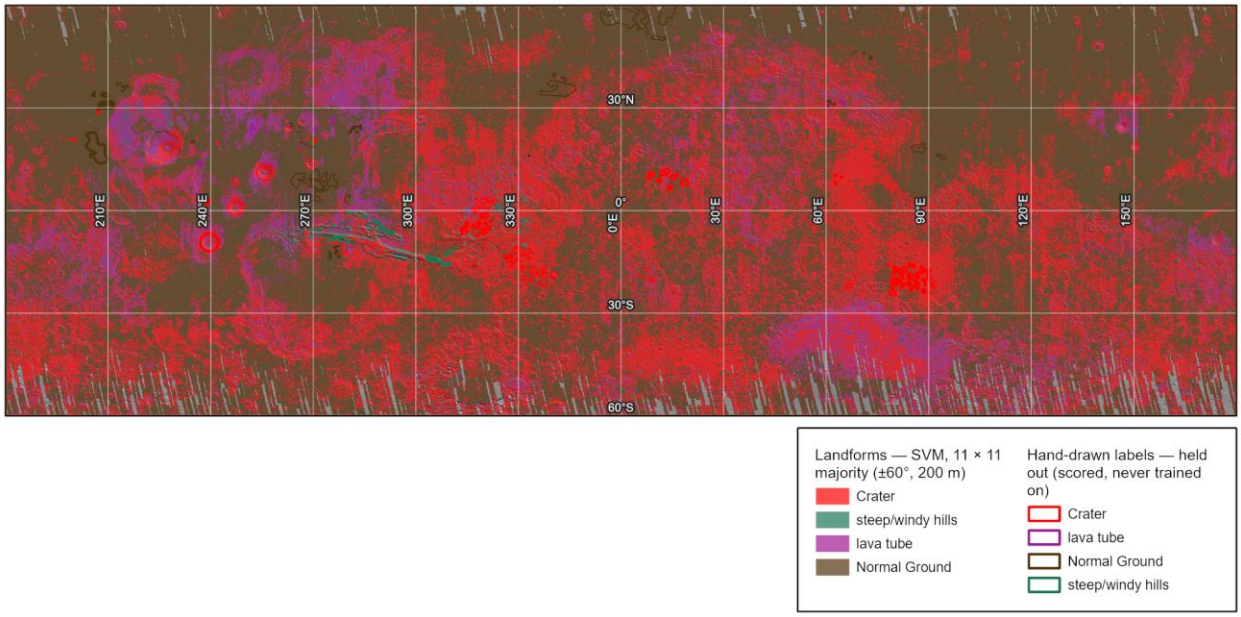
- Flood lava once mapped as a water channel; the USGS map calls it Late Amazonian volcanic (lAv)
- Built the same way as Ius: 3,283 channel candidates, 409 of them rock-floored
- Catalogued craters: 1,122 per million km² on lAv, against 1,760 and 2,183 on the older units
- The diurnal-contrast index does not track calibrated TES inertia here ($r \approx 0$); Viking colour does (-0.55)

Figure 11. Checking the first two classifications



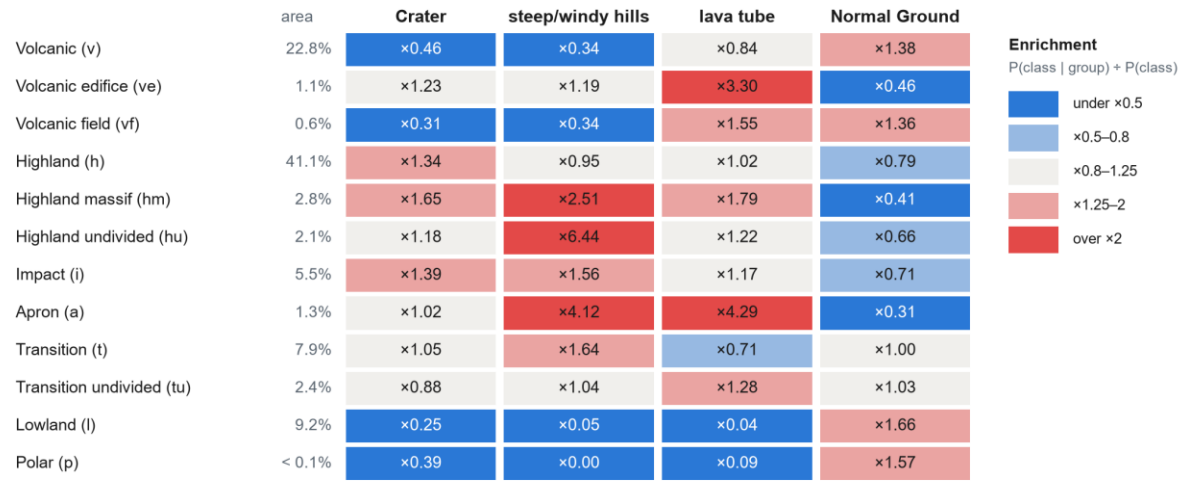
- The 29 Sep map follows its 4096-px processing tiles instead of the terrain
- The 30 Sep map is mostly elevation: elevation alone reproduces 62% of it
- Both problems led to the corrected stack, with no elevation band and slope in degrees

Figure 12. Landforms at ±60°, scored on labels the model never saw



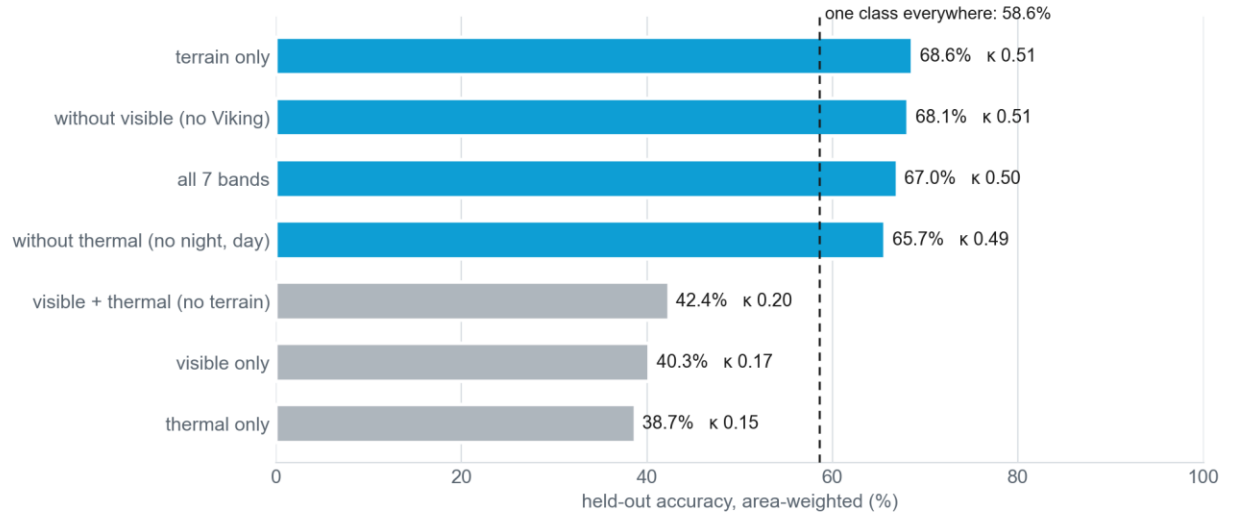
- 512 hand-drawn polygons in whole 15° blocks: 329 train the SVM, 183 are held out
- Published map, 200 m, 11 × 11 majority (2.2 km): 73.8%, κ 0.58 held out; one class everywhere: 58.6%
- Halving the cell bought no accuracy: 69.9% raw at 200 m, 70.5% at 400 m
- Normal Ground and cratered terrain hold up; lava tube does not (2% user's accuracy)

Figure 13. The landform classes against the USGS geologic map



- Each cell: how much more often a class occurs in a unit group than across ±60° (×1 = no link)
- "Lava tube" gathers on volcano flanks and aprons (×3.30, ×4.29), not on volcanic plains (×0.84)
- 76% of the volcanic plains come out Normal Ground
- The classes follow terrain, so lava flows are taken from the geologic map

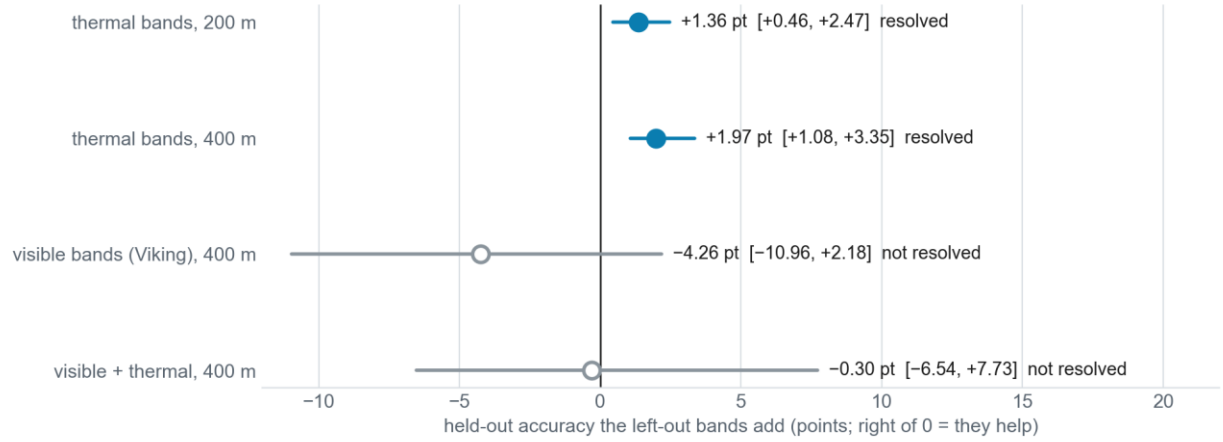
Figure 14. Does thermal infrared improve the classification?



Cyan: band sets that include slope and relief. Thermal bands add +1.3 pt (95% CI +0.6 to +2.2), κ +0.013. Gaussian maximum likelihood, 183 held-out polygons, 400 m.

- Slope and relief alone: 68.6%, κ 0.51 held out; all seven bands: 67.0%, κ 0.50
- The thermal bands add +1.3 points of overall accuracy (95% interval +0.6 to +2.2), small but clear of zero
- Without terrain, visible and thermal score below one class everywhere (58.6%)
- The labels are landforms, so terrain decides them; testing thermal IR properly needs material labels

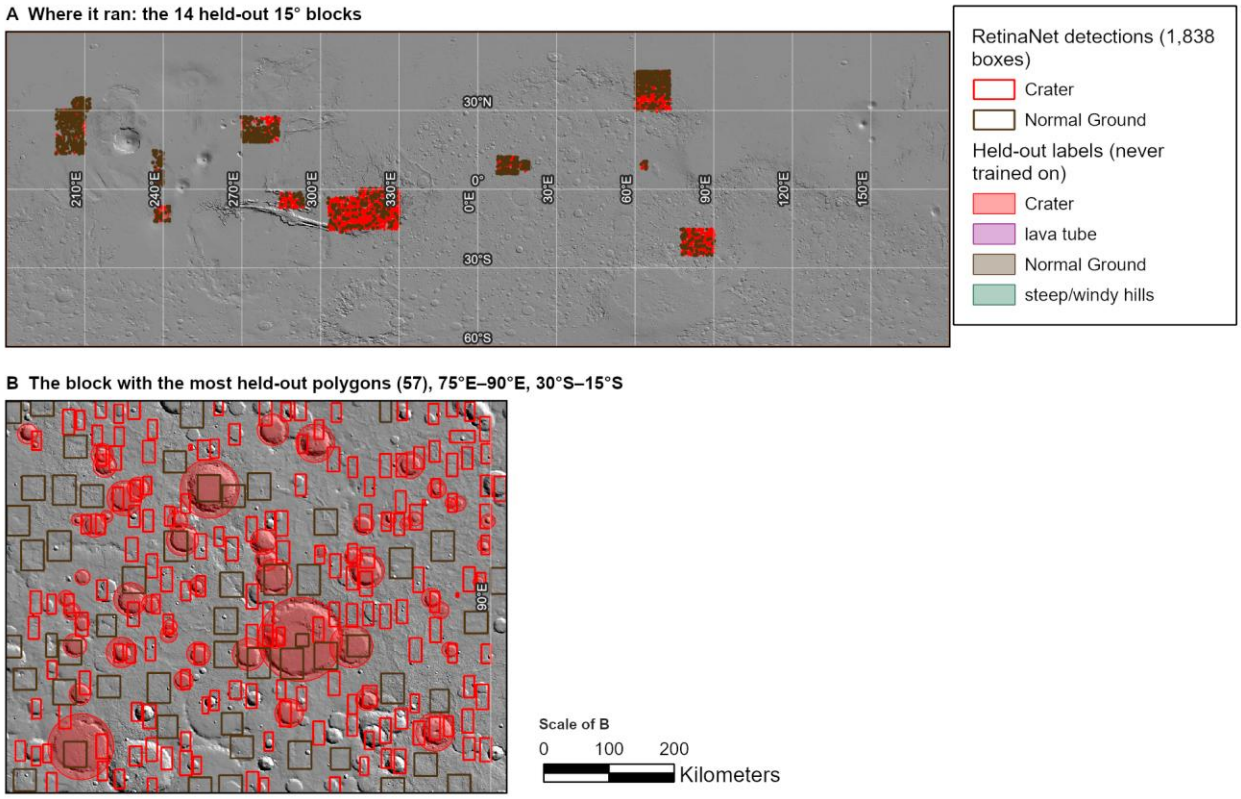
Figure 15. The thermal gain, checked with ArcGIS Pro's own SVM



ArcGIS Pro SVM over ±60°, the same split and settings for every band set. Dot: all bands minus the set; line: paired-bootstrap 95% interval, 183 held-out polygons, 1,000 resamples.

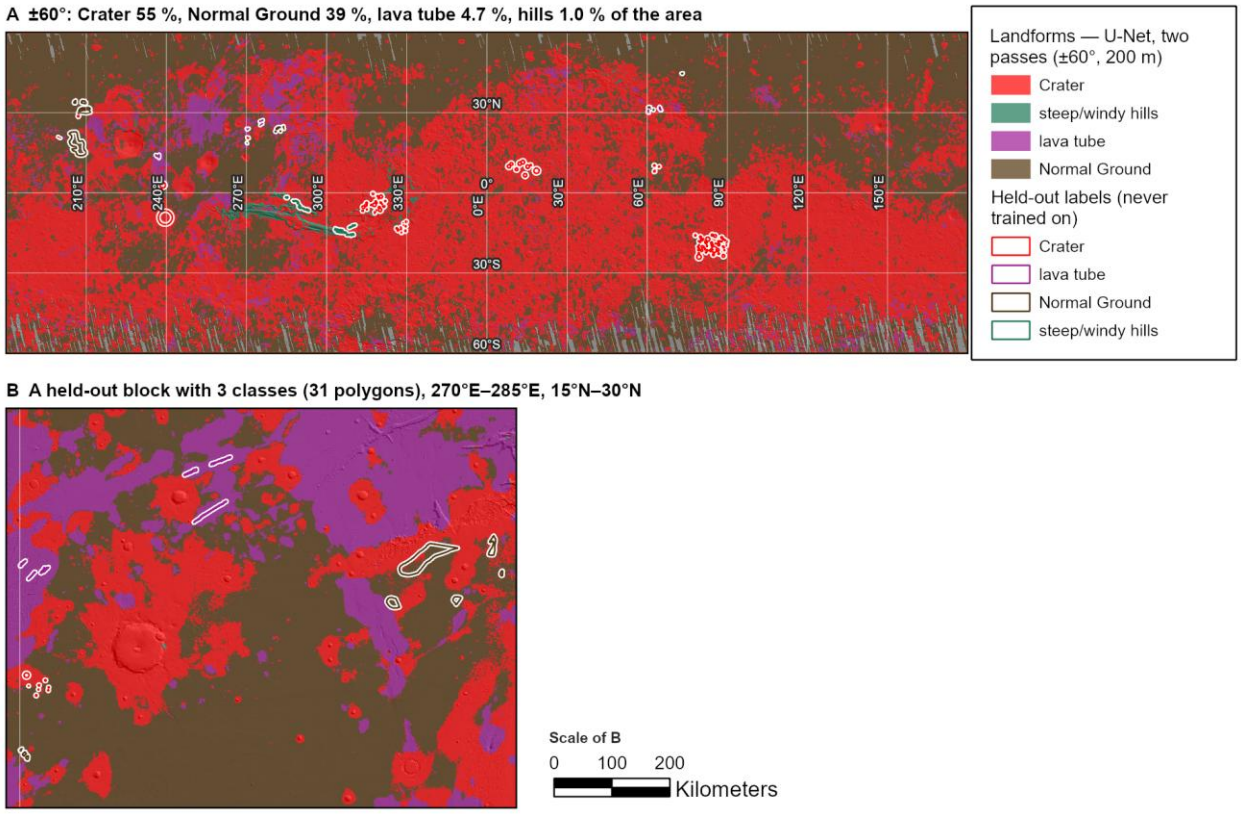
- Pro's SVM retrained over all of ±60° with the published settings, leaving out one band group at a time
- Thermal adds +1.36 points of overall accuracy at 200 m (+0.46 to +2.47) and +1.97 at 400 m (+1.08 to +3.35)
- It helps craters and hills and hurts lava tube, so with every class weighted equally (balanced accuracy) the gain disappears: −0.31 points at 200 m, +0.07 at 400 m
- Leaving out visible, or both groups, ties the score; terrain alone ties it too but gets only 34 of 124 crater polygons right

Figure 16. Deep learning, 1: a detector that draws boxes



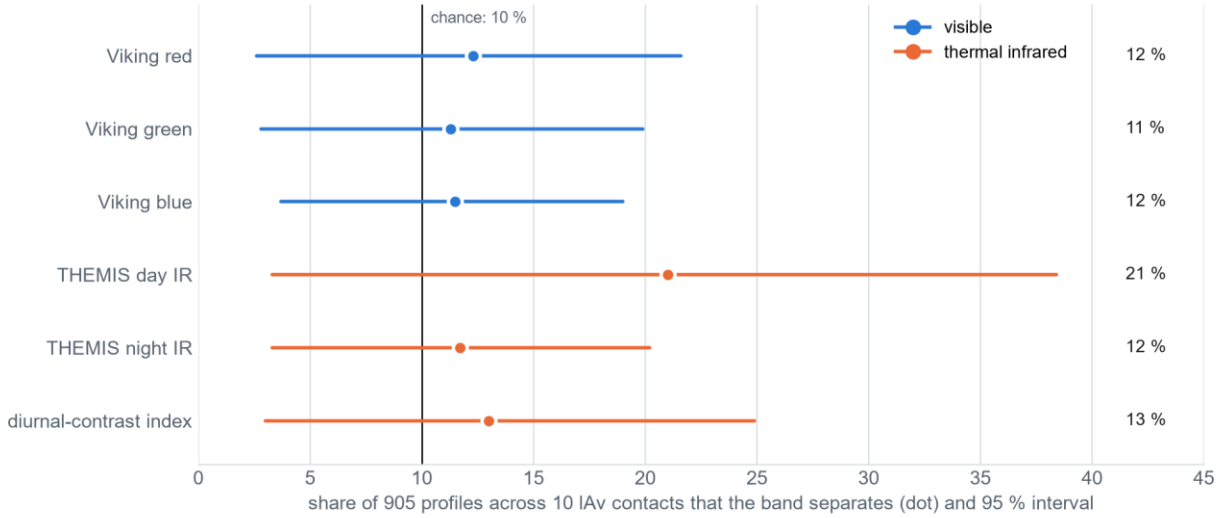
- RetinaNet on the 7-band stack, trained on the training blocks only, scored on the 183 held-out polygons
- Crater: 41.9% of polygons found, 87.6% of boxes right, against 71.1% for always naming Crater
- Normal Ground: 22.7% found, 56.9% right; hills and lava tube never named
- Retrained with the backbone unfrozen for 30 epochs, it names hills but finds only 4.5% of Normal Ground, so it is no better

Figure 17. Deep learning, 2: a U-Net maps terrain provinces



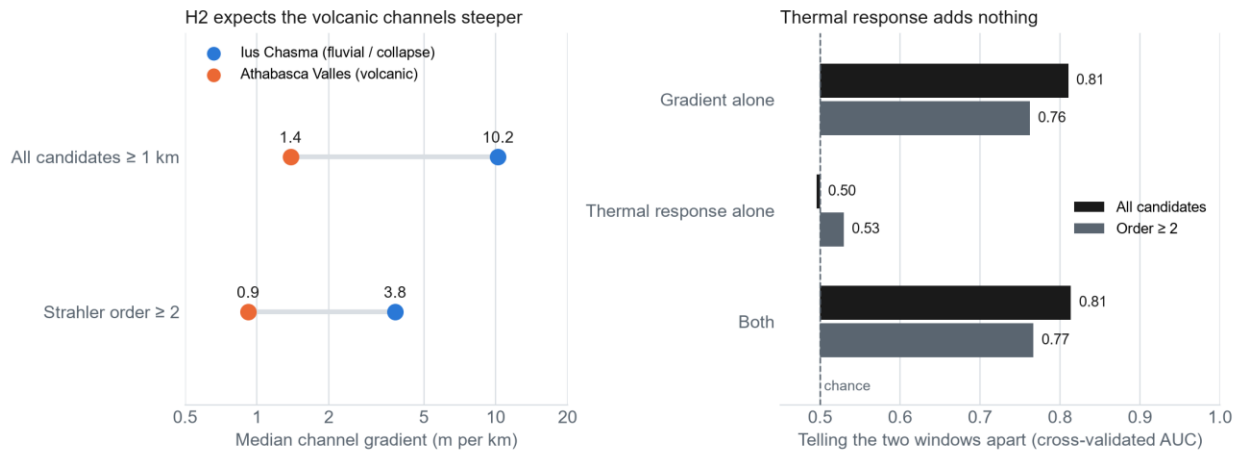
- A U-Net pixel classifier on the same stack and labels, over all of ±60°; a second pass, offset 128 px, removes its tile seams
- Held out: 98.9%, κ 0.980, against the published SVM map's 73.8%, κ 0.583, under the same scorer
- Lava tube becomes usable: 77.1% user's accuracy, against 2.4% for the SVM
- It maps provinces rather than single landforms: its 'Crater' covers 82.6% of the ring around named craters. The SVM map stays the published one

Figure 18. Do the infrared mosaics show lava-flow edges Viking misses? (H1)



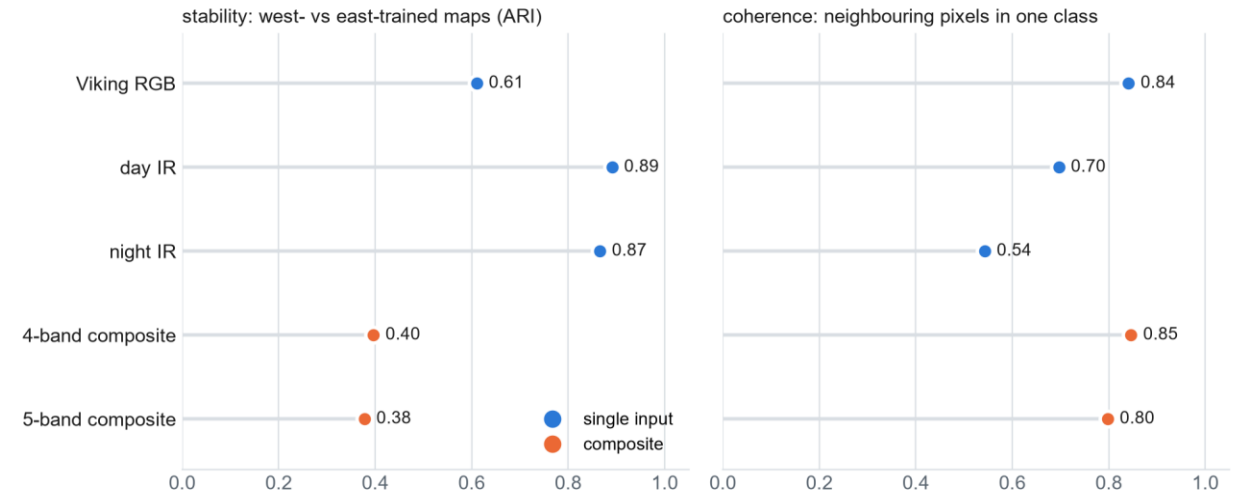
- 905 profiles across 10 of the geologic map's lava contacts at Athabasca: does each band tell the units apart?
- Every band does only slightly better than chance (10%): Viking red 12.3%, day IR 21.0%
- Day IR leans toward H1, but the intervals overlap (difference +8.7 points, −1.7 to +18.1)
- Contacts on a 1:20 M map are kilometres wide; H1 is retested on margins digitised at 100 m

Figure 19. Do volcanic channels differ from fluvial ones? (H2)



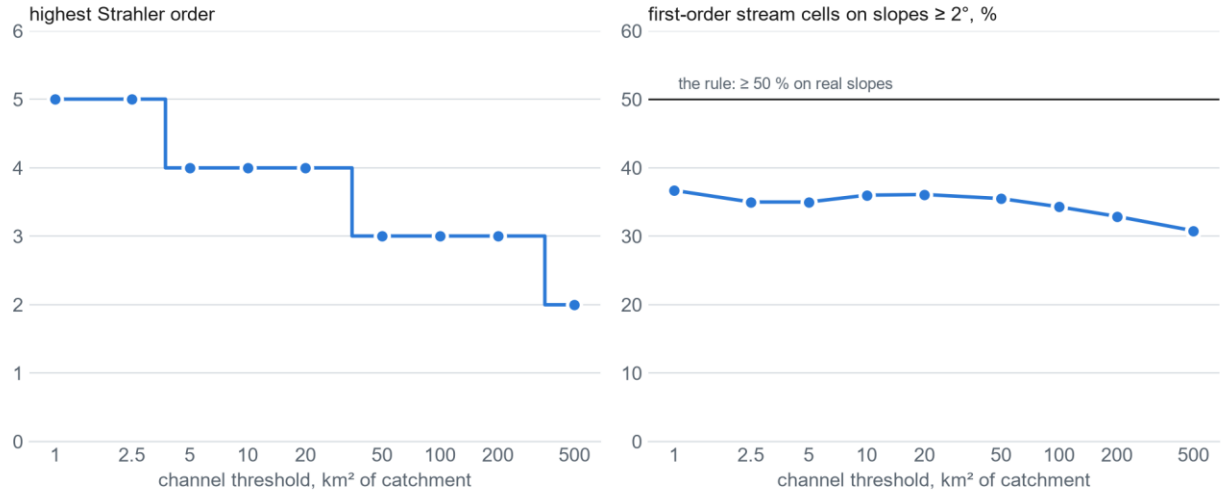
- Channel candidates, Ius Chasma (fluvial) against Athabasca Valles (volcanic): 2,588 and 3,272
- H2 expected the volcanic channels to be steeper; they are shallower (median 1.4 against 10.2 m per km)
- Thermal response tells the two apart no better than chance (AUC 0.50)
- Not supported at stage 1: the window sets the gradient, not the origin. Reviewed channels retest it

Figure 20. Is the composite more stable than a single band? (H3)



- Iso Cluster trained on the west and the east half of Ius, compared over the whole window
- The composites are the least reproducible inputs: 0.40 and 0.38, against 0.61–0.89 for single bands
- They give the smoothest maps, only just ahead of Viking (0.85 against 0.84)
- H3 is not supported

Figure 21. How many tributary orders does 100 m resolve? (Q5)



- Channel thresholds from 1 to 500 km² of catchment at Ius Chasma
- The highest stream order follows the threshold: 5 at 1 km², 3 at 50 km², 2 at 500 km²
- Only 30.8–36.7% of first-order streams lie on slopes ≥ 2°, at every threshold
- The order count reflects the threshold chosen, so the DEM cannot answer Q5

7. Project Issues and Hurdles

- **The thermal signal is weak where it can be checked.** The THEMIS mosaics follow calibrated TES thermal inertia only weakly inside 15° blocks ($|r| \approx 0.1$) and not at all across the planet; Viking colour follows it (r −0.31 to −0.36). In the classification the thermal bands add 1.3–2.0 points of overall accuracy, with two classifiers, but none by balanced accuracy.
- **The landform classes are terrain classes.** Slope and relief alone score as well as all seven bands. On the geologic map the classes sit where terrain puts them, and "lava tube" lands on volcano flanks rather than lava plains.
- **Five "Crater" labels are volcanoes.** One label, on Arsia Mons, is 31% of the held-out Crater area; the seven-class labels make them volcanic edifices.
- **Lava flows come from the geologic map.** These bands cannot tell volcanic plains from other plains, and a fifth "volcanic" class scored below chance. Lava flows are taken from the USGS map's volcanic units.
- **Closed depressions are not a crater count.** Of the candidates ≥ 1 km, 11–13% in the type areas and 15.3% across ±60° are catalogued craters; from 5 km up most are (62.2%). The Robbins catalogue (385,049 craters) is the crater reference.
- **Four hypotheses tested, none supported.** No band separates the geologic map's lava contacts much better than chance, though the map is too coarse to settle H1; the volcanic channels are shallower, and thermal response separates nothing (H2, stage 1); the composite's classes are the least reproducible input (H3); no channel threshold keeps first-order streams on real slopes (Q5).

- **The review has only begun.** 549 of 9,287 candidates reviewed, all at Ius (precision 54% on steep channels, 7% on depressions); H1 and H2 wait.
- **428.7 GB on one USB drive.** Reads cap near 112 MB/s, and the drive has dropped off mid-write, so big outputs are built on internal disk first.
- **Solved since September.** Three coordinate frames, the z-factor defect, four failed global composites, the missing night mosaic, a defective DEM column at 180°.

8. Next Steps

- Continue the review: the rest of Ius Chasma, then Athabasca Valles; draw breached craters and lava margins
- Score the candidates against the reviewed features on the validation sheet; retest H1 on digitised margins and H2 on reviewed channels with their origin set
- Seven classes: the 971 reviewed labels are applied and the SVM and U-Net are rerunning; score both maps on the same held-out blocks
- Deep learning: set aside validation blocks before any further tuning, so the held-out blocks stay a clean test

When	What
8–26 Oct	Pyramids, ±60° crater and channel pass, 200 m map, T1 in Pro, the deep-learning detector and U-Net; seven classes proposed
26 Oct	Interim data cut; later results go to the final report
27 Oct – 22 Nov	Review and digitising (begun 26 Oct); the seven-class maps; H1 and H2 on digitised features; data freeze
10 Nov	Interim presentation
23 Nov – 4 Dec	Final report and presentation
8 Dec	Final due